

29 Jan 2026

FINDVAL (x, A)

↙ search value
↙ array

```

1:  $i \leftarrow 1$ 
2: while  $i \leq |A|$ 
3:   if  $x = A[i]$ 
4:     return TRUE
5:   end if
6:    $i++$ 
7: end while
8: return FALSE
  
```

↑ Linear search through array

runtime (worst-case)

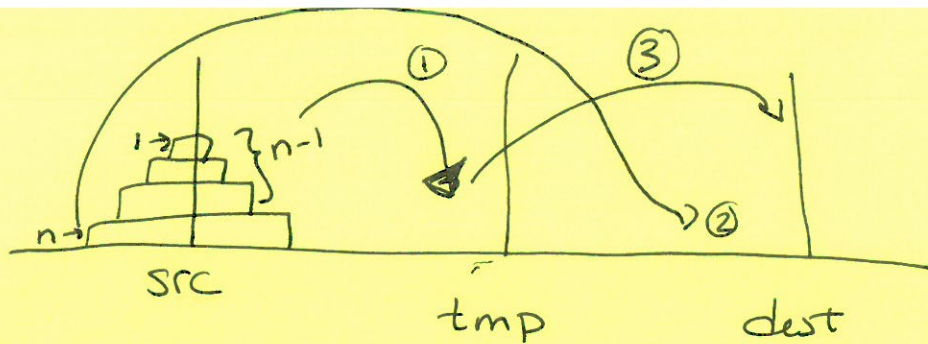
```

1:  $\Theta(1)$ 
2:
3: } For loop
4: } ( $\# \text{ times}$ )  $\cdot$  ( $\text{cost per time}$ )
5: }  $\Theta(|A|) \cdot \begin{cases} 2: \Theta(1) \\ 3: \Theta(1) \\ 4: \Theta(1) \\ 5: \Theta(1) \\ 6: \Theta(1) \end{cases} = \Theta(|A|)$ 
6:
7:
8:
  
```

Letting $n = |A|$,

$$\Theta(1) + \Theta(n \cdot 1)$$

$$= \Theta(1 + n) = \boxed{\Theta(n)}$$



HANOI ($n, \text{src}, \text{dest}, \text{tmp}$)

1: if $n > 0$

2: HANOI ($n-1, \text{src}, \text{tmp}, \text{dest}$)

3: ~~move~~ move $n^{\text{th largest}}$ disk from src to dest

4: HANOI ($n-1, \text{tmp}, \text{dest}, \text{src}$)

5: endif

6: return

a function $T: \mathbb{N} \rightarrow \mathbb{R}$

Define $T(N)$ to be the runtime of HANOI when $n = N$.

$$T(N) = \underbrace{\Theta(1)}_{\text{Line 1}} + \underbrace{T(N-1)}_{\text{Line 2}} + \underbrace{\Theta(1)}_{\text{Line 3}} + \underbrace{T(N-1)}_{\text{Line 4}} + \underbrace{\Theta(1)}_{\text{Line 5/6}}$$

$$= 2T(N-1) + \Theta(1)$$

(In CS, we always assume the base case is const: $T(1) = 1$)

So, we write:

$$\boxed{T(N) = 2T(N-1) + \Theta(1)} \in \Theta(2^n)$$

EXponential!

Proof: Uses induction, can be done on
(a) exact formula or (b) use c, no in def. of Θ (2)

$$T(n) = \begin{cases} 1, & n=1 \\ T(n-1) + n, & n > 1 \end{cases} \quad (*)$$

Questions:

(a) Name an algo that has this recurrence. (seed!)

Selection sort
(worst-case runtime for Quicksort)

(b) Find the closed form of it. asymptotic form?

[often, educated guess + check]

$$T(n) = \frac{n(n+1)}{2} \quad T(n) \in \Theta(n^2)$$

(c) Expand it out + write as a Σ

$$T(n) = \sum_{i=1}^n i = \frac{n(n+1)}{2}$$

(d) How would we typically write this as computer scientists?

$$T(n) = T(n-1) + \Theta(n)$$

(b')
(c') Try it for a few small #s.

$$\begin{aligned} T(3) &= T(2) + 3 && \text{by def'n of case } n > 1 \\ &= (T(1) + 2) + 3 && \text{"} \\ &= ((1) + 2) + 3 && \text{by def'n of base case} \\ &= \sum_{i=1}^3 i \end{aligned}$$

a glance into the induction.

Let $T(n)$ be defined as $(*)$.

~~Let~~ Claim: $T(n) = \frac{n(n+1)}{2}$

Proof: Base case: Let $n=1$.

$$T(1) = 1.$$

$$\text{RHS: } \frac{n(n+1)}{2} = \frac{1 \cdot 2}{2} = 1 \quad \checkmark$$

Let $k \geq 1, k \in \mathbb{N}$

assume that

$$T(k) = \frac{k(k+1)}{2}$$

closely related!!

((WTS: "This" holds for $k+1$
 $\hookrightarrow T(k+1) = \frac{(k+1)((k+1)+1)}{2}$))

Because $k \geq 1$, we know $k+1 \geq 2$.

$$T(\underline{k+1}) = T(\underline{(k+1)-1}) + \underline{(k+1)} \quad \text{by def'n of } T$$

$$= T(k) + (k+1)$$

$$= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} \quad \text{by i. A.}$$

$$= \frac{(k+2)(k+1)}{2}$$

$$= \frac{(k+1)((k+1)+1)}{2}$$

